Chapter 5 Experimental Design

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Statistical Decision Making (45-752)

[ ]:
```
import tprstats
import pandas
import matplotlib.pyplot as plt
import numpy as np
from statsmodels import stats
from scipy.stats import t

# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f
```

[ ]: '%.4f'

1 Experimental Design

1.0.1 Learning objectives:

After completing this chapter, you will be able to:

- Explain the four possible outcomes of a hypothesis test.
- Define and calculate power of a hypothesis test about a population mean.
- Calculate sample size to test a hypothesis about a population mean.
- Calculate power of a test of a hypothesis about a difference in population means.
- Calculate sample size to test a hypothesis about a difference in population means.
- Use regression to test a hypothesis about a difference in population means.

1.1 Design of Experiments

In Chapter 3, we defined a **Randomized Control Trial (RCT)** and detailed the rationale for using RCTs to evaluate alternatives. In recent years, experiments have come to be used much more widely in managerial decision making. Hence, it is important for you to know how experiments are designed. It is also important for to know the decisions that you, as a manager, must make in supervising the design of an experiment.

Experimental design is best understood in the context of applications.

- We will study design of experiments in which the goal is to test a hypothesis about a population mean.
- We then study design of experiments to test a hypothesis about a difference in population means.

1.2 Four Possible Outcomes of a Hypothesis Test

You are, of course, familiar with the format of a hypothesis test.

Null Hypothesis: The hypothesis that is accepted unless sufficient evidence to the contrary is obtained.

Alternative Hypothesis: The hypothesis that is tested against the null hypothesis. The alternative hypothesis is accepted if sufficient evidence against the null is found.

A hypothesis test has four possible outcomes:

- We accept the null hypothesis and the null hypothesis is true.
- We accept the null hypothesis and the null hypothesis is false. This is called a **Type II Error**.
- We reject the null hypothesis and the null hypothesis is false.
- We reject the null hypothesis and the null hypothesis is true. This is called a **Type I Error**.

The different outcomes are illustrated in the following table.

		Null Hypothesis	
		True	False
Decision	Accept	Correct Decision Probability= (1-α)	Type II Error Probability = β
	Reject	Type I Error Probability= α Called Significance Level	Correct Decision Probability= (1-β) Called Power

We would like to accept a null hypothesis when the null hypothesis is true and reject a null hypothesis when the null hypothesis is false. However, as noted above, there are two types of errors that may occur in hypothesis testing. A **Type I Error** occurs when we reject a null hypothesis that is true. A **Type II Error** occurs when we accept a null hypothesis that is false.

From your work with hypothesis testing, you are familiar with the two items under **True** in the diagram. We choose a significance level α, normally a relatively small value such as $\alpha = .05$. This is the probability that we will make a Type I error and reject a true null hypothesis. The probability that we will accept a true null hypothesis is then relatively high, $1 - \alpha$. Viewing the two items under **False** in the diagram, we see that the probability of a Type II error is denoted β.[1] Of course, we would like this probability to be small. You see as well from the diagram that the probability of rejecting a false null hypothesis is denoted $1 - \beta$. This is Power. We would like Power to be high.

[1] This is a standard use of symbol β in experimental design. It should not be confused with use of beta's as regression coefficients

1.3 Application: Test a Hypothesis about a Population Mean

Automotive Testing Service (ATS) is a company that tests fuel efficiency claims made by automobile manufacturers. To test fuel economy claims, ATS chooses a route in a metropolitan area that is typical of normal city driving. ATS then selects a random sample of drivers from a pool of volunteer drivers. Each driver is assigned a randomly chosen vehicle from the manufacturer's inventory of the vehicle type being tested. The driver drives the vehicle on the route laid out by ATS, and ATS measures the fuel economy obtained by each of the vehicles. ATS then determines whether the mean miles per gallon from the sample, $\overline{X}$, is or is not consistent with the manufacturer's claim.

1.4 Evaluate Manufacturer's Claim about Fuel Economy of A Car

An automobile manufacturer claims its new hybrid SUV will obtain 40 miles per gallon in normal city driving. Automobile Testing Service (ATS) will evaluate their claim as follows:

- ATS will select a random sample of n=50 drivers.
- Each driver will drive a randomly chosen SUV from the manufacturer's inventory.
- ATS will assess whether the mean mpg from the sample, $\overline{X}$, is consistent with the manufacturer's claim.

Null Hypothesis: SUV obtains 40 mpg
Alternative Hypothesis: SUV does not obtain 40 mpg

1.5 Calculate Power of The Test

Before conducting its evaluation, ATS wants us to calculate the power of the test it is planning to conduct.

Definition: **Power** is the probability of rejecting a false null hypothesis.

We will do the power calculation for ATS for its planned experiment.

Information and notation for the power calculation for the planned experiment:

- μ is the unknown true MPG of this SUV
- $\mu_0 = 40$ is the MPG claimed by the manufacturer
- Sample size n=50
- Based on prior experience, ATS expects $\sigma = 5$ for this SUV
- ATS chooses significance level $\alpha = .05$
- ATS asks us to calculate the power of its test if the manufacturer's claim is wrong by more than 2 mpg. This is called **effect size**. ATS considers a manufacturer's claim to be acceptable if the actual MPG is within 2 miles per gallon of the manufacturers claim. The choice of effect size is a key managerial decision that you must make when designing an experiment.

To calculate power, it is necessary to specify the standard deviation, σ of mpg for the type of vehicle being tested. Note that σ is needed for the power calculation before the experiment is conducted. Hence, ATS relies on its extensive knowledge of automobile testing to make an informed judgment about the value of σ that is likely to be appropriate for the SUV being tested. When an experiment is being designed, the individuals designing the experiment will usually have expertise to enable them to make such an informed judgment about σ.

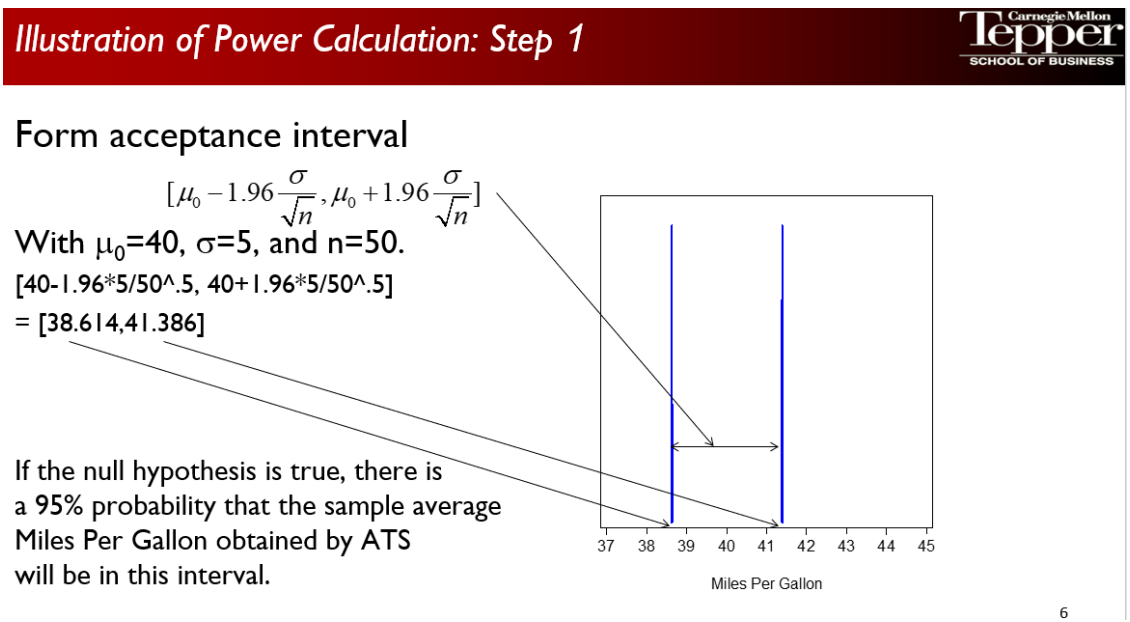
There is random variation in fuel economy due to differences in drivers, variation in traffic conditions encountered by a given driver, and perhaps some differences in across the vehicles in the sample

being tested. Hence, ATS needs to choose a large enough sample to judge the validity of the manufacturer's fuel economy claim. To decide how many vehicles must be tested, ATS must make tradeoffs. To understand these tradeoffs, let's think about the possible outcomes of a hypothesis test and develop some terminology that will be useful as we go forward with experimental design.

1.6 Graphical Portrayal of Calculation of Power

1.6.1 Calculating Power: Step 1

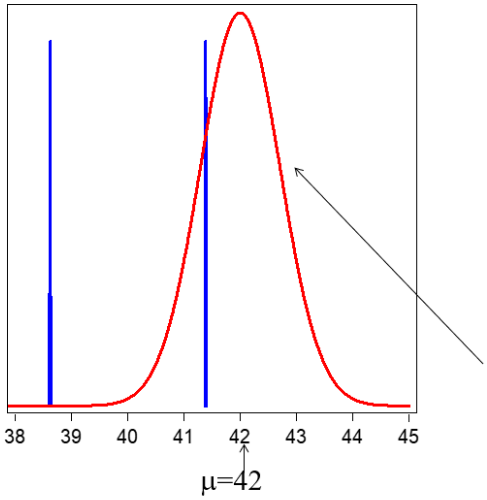
The first step in a power calculation is to form the acceptance interval. This will be familiar to you because we used acceptance intervals when we studied statistical quality control. The manufacturer claims the population mean for this SUV is 40mpg. The acceptance interval is displayed below. We see that ATS will accept the manufacturer's claim if the average mpg for the n vehicles tested falls in the interval [38.61,41.37]. Note where μ_0, σ, and sample size n are used in the calculation of this acceptance interval.



1.6.2 Calculating Power: Step 2

ATS will accept the manufacturer's claim if the sample mean from the experiment falls within +/- 2mpg of the manufacturer's claim. Hence, ATS asks us to calculate the power of the test if the manufacturer's claim is wrong by 2 mpg. Suppose the manufacturer's claim is off by 2 mpg and the actual mpg=42. The red bell-shaped curve is a normal distribution with $\mu = 42$ and $\sigma = 5$. Thus, this normal distribution is centered 2 mpg above the manufactures claim, and has standard deviation which ATS believes is appropriate for the SUV been tested. The area under this normal distribution that lies outside the acceptance interval is the probability that the null hypothesis will be rejected if true mpg is 42, i.e. power.

Illustration of Power Calculation: Step 2

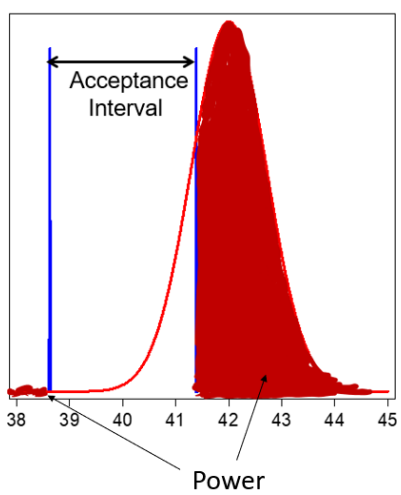


- ATS asks us to calculate power if true MPG differs from manufacturer's claim by 2 MPG.
- The sample mean $\bar{X}$ is a random variable.
- ATS will obtain one sample mean $\bar{X}$ when it conducts is evaluation.
- Here is the plot of the distribution of possible values of $\bar{X}$ ATS might obtain if the true MPG is μ=42 and the standard deviation is 5.

7

The display below shades in the area corresponding to power, with further explanation provided beside the diagram.

Illustration of Power Calculation Continued



The acceptance interval is centered on the null value μ_0=40.

The red distribution curve is the distribution of $\bar{X}$ if μ=42 (and σ=5).

The red area under the distribution curve and outside the acceptance interval is the probability you reject the null hypothesis if the true MPG is μ=42.

Hence, the sum of the red areas is the Power of the test when the true mean mpg is μ=42 and the null hypothesis is μ_0=40.

We can use the normal distribution to calculate power because n is large enough to rely on the central limit theorem.

Optional Detail for Power Calculation

1.7 Calculations for Testing a Hypothesis About a Population Mean

1.7.1 Using Python to Calculate Power Given Sample Size

Suppose ATS wishes us to investigate power if ATS chooses a sample size of 50. The commands calculate power for an experiment with effect size=2, anticipated standard deviation=5, sample size = 50, and significance level .05. Be sure you see where each number in the commands is obtained from the above discussion. If you were evaluating a different experiment, you might need to change one or more of these four numbers in the commands. We see that we obtain power of .79. Suppose a colleague asks you what this means. How would you answer?

[72]:
```
effect_size = 2
stdev = 5
standardized_effect_size = effect_size / stdev
stats.power.tt_solve_power(effect_size=standardized_effect_size, nobs=50,␣
↪alpha=0.05)
```

[72]:
```
np.float64(0.7917871892678123)
```

1.7.2 Using Python to Calculate Sample Size Given Power

Let's calculate sample size for the Automotive Testing Service (ATS) application. To calculate sample size, it is necessary to specify the power to be achieved by the test. As we will see, the higher the desired power, the larger the sample size that will be required. Suppose ATS asks us to calculate sample size to achieve power=.9. In the code below, the first two lines set the effect size and the standard deviation. The third line calculates the standardized effect size. Then the tt_solve_power command calculates the sample size we need to obtain power=.9. After running the code below, we see that we need a sample size of 68.

[73]:
```
effect_size = 2
stdev = 5
standardized_effect_size = effect_size / stdev
stats.power.tt_solve_power(effect_size=standardized_effect_size, power=0.9,␣
↪alpha=0.05)
```

[73]:
```
67.6214
```

1.7.3 Conduct Hypothesis Test When Data Have Been Collected

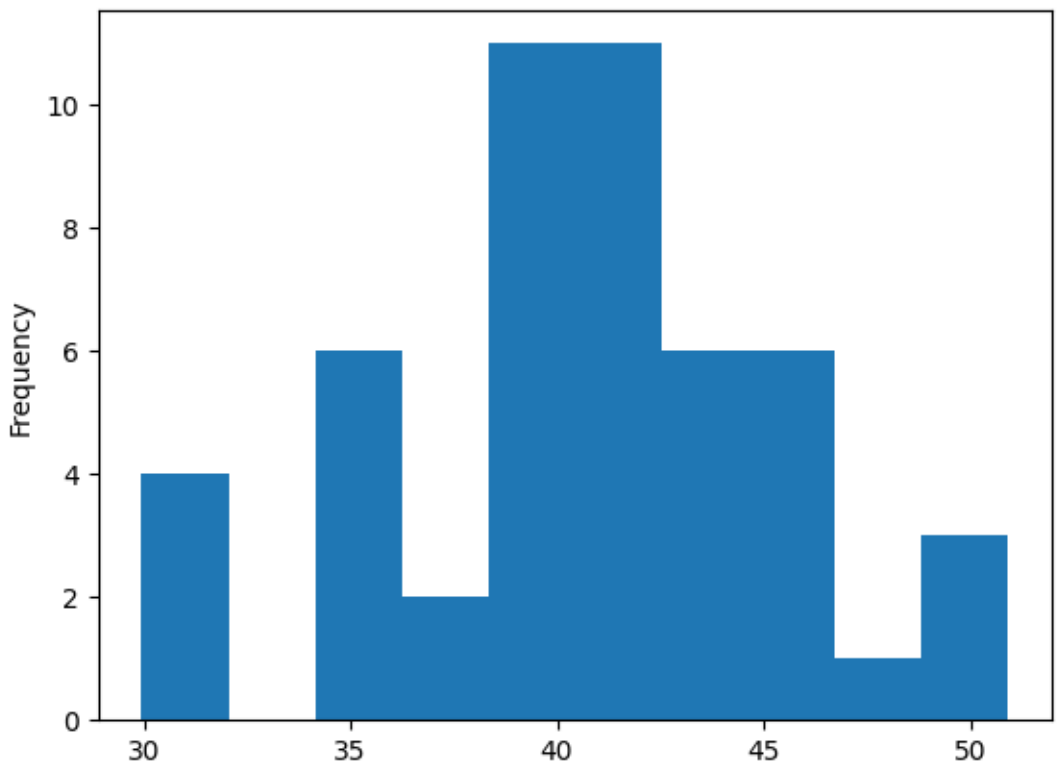
The results of the ATS experiment are in the Excel file named SUV.
Import SUV.xlsx

[74]:
```
SUV = pandas.read_excel("data/SUV.xlsx")
```

Before testing the hypothesis, let's view a histogram of the miles per gallon data collected in the experiment. We see no extreme outliers or other cause for concern.

[75]:
```
SUV["mpg"].plot.hist()
```

[75]:
```
<Axes: ylabel='Frequency'>
```



Use the data to test the null hypothesis μ_0 that population mean mpg is 40. We obtain a p-value = .212. This is greater than our chosen significance level of .05. Hence, ATS does not reject the null hypothesis that the population mpg for its SUV is 40. Note that the sample mean is actually slightly above the manufacturer's claim. This data set is from 50 vehicles tested by ATS. As we noted above, ATS would need to test 58 vehicles to obtain power =.85 and 68 vehicles to obtain power =.90.

[76]:
```
tprstats.test_population_mean(SUV["mpg"], 40)
```

[76]:
```
t_statistic                  1.264470
p_value                      0.212043
confidence_interval_lower   39.504548
confidence_interval_upper   42.177052
sample_mean                 40.840800
dtype: float64
```

By the way, when the experiment is completed, ATS might reject the manufacturer's claim because the actual fuel efficiency is higher than the claimed fuel efficiency. Of course, such a result would be favorable to the manufacturer. ATS would give the manufacturer a positive review because the performance of the SUV in the experimental evaluation exceeded the manufacturer's claim.

An additional item to note is that effect size is sometimes expressed in standard deviations. In the above ATS application, σ equals 5. Hence, the effect size of 2 might be expressed as 2/5 or .4 standard deviations, so be attentive to how effect size is defined if you are reading about findings

of an experiment.

1.8 Building Intuition About Effect Size

We introduced the concept of Effect Size in the above analysis. We now consider examples to build intuition about Effect Size using the ATS experiment to evaluate SUV mileage.

Suppose the true mpg of the SUV being evaluated is 15mpg.

Would it be hard to reject the manufacturer's claim of 40 mpg?

- No, it would be easy.

Why would it be easy?

- A relatively small sample would be sufficient to reject the claim of 40 mpg because a difference (effect size) of 25 mpg between null (40) and true (15) would be easy to detect.

Suppose the true mpg of the SUV is 39.5 mpg.

Would it be hard to reject the manufacturer's claim of 40 mpg?

- Yes.

Why would it be hard?

- A very large sample would be required to reject the claim of 40 mpg because such a small difference (effect size) of 0.5 mpg between null (40) and true (39.5) would be very difficult to detect.

Suppose the true mpg of the SUV is 15 mpg.

Would it be important for Automotive Testing Service to reject the manufacturer's claim of 40 mpg?

- This would obviously be important!

Suppose the true mpg of the SUV is 39.5.

Would it be important for ATS to reject the manufacturer's claim of 40 mpg?

- Probably not. Such a small deviation from the actual mpg is not of much concern.

1.9 Key decisions you need to make in managing an experiment

1. **Effect Size:** How small a difference you will seek to detect between the null and alternative with your experiment?
2. **Power:** How large a probability you will require to reject a false null hypothesis?
3. **Significance level:** How small a probability will you adopt for rejecting a null hypothesis when it is true?
4. **Standard deviation of the variable being tested.** For example, in the ATS experiment, it is necessary to adopt a value for the standard deviation of miles per gallon for the vehicle.

The first two questions above are where managerial decision making plays a particularly important role. For the third, it is quite common to choose $\alpha = .05$ though there may be circumstances where a smaller α would be chosen. An α greater than .05 would be relatively uncommon. For the fourth,

you or your company would typically have prior experience to make the judgment about standard deviation.

Reducing effect size increases the cost of an experiment because more subjects are required. Similarly, increasing power increases the cost of an experiment because more subjects are required. Hence, you need to trade off cost versus effect size and power when you manage an experiment.

In our ATS application, the manager decided that an Effect Size of 2 mpg would be sufficiently large that it would be important to detect such a deviation from the manager's claim. It is often the case that power is chosen to be on the order of .8 or .85. For practice, complete Exercise 1. Suppose the manager of ATS adopts a standard of .85 for its testing. Please confirm, by modifying the above command, that 58 vehicles would need to be tested.

1.10 Testing A Hypothesis about A Difference In Population Means

When a pharmaceutical company creates a new medication for treating an illness, it must validate the safety and effectiveness of the medication. This is done by experiments that compare the efficacy of the proposed new medication to the best existing medication for treating the illness. The US Food and Drug Administration (FDA) has legal responsibility for specifying the requirements that must be met before a proposed new medication can be prescribed by physicians. The FDA has promulgated stringent requirements for experiments testing a new medication. Visit the following web page to learn about the approval process:

https://www.fda.gov/drugs/development-approval-process-drugs

In recent years, A/B testing has become much more widely used in managerial practice. We will now see how to design an experiment to test a difference in population means. We previously covered in Chapter 3 testing of a difference in populations means. We begin with a brief recap and then turn to designing an experiment to test a difference in populations means.

1.10.1 The t-statistic for The Test of Differences in Means

Let X and Y to denote the two alternatives, with μ_x denoting the mean of population X and μ_y the mean of population Y. We wish to test a null hypothesis about a difference in population means: $(\mu_{x,0} - \mu_{y,0})$. If we have a simple random sample of size n_x from population X and a simple random sample of size n_y from population Y, the t-statistic for the test of differences in population means is:

$$t = \frac{(\overline{X} - \overline{Y}) - (\mu_{x,0} - \mu_{y,0})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} + \frac{s_{\overline{y}}^2}{ny}}}$$

where $s_{\overline{x}}^2$ and $s_{\overline{y}}^2$ are the sample variances:

$$s_{\overline{x}}^2 = \frac{\sum_{i=1}^{n_x}(X_i - \overline{X})^2}{n_x - 1} \text{ and } s_{\overline{y}}^2 = \frac{\sum_{i=1}^{n_y}(Y_i - \overline{Y})^2}{n_y - 1}$$

Often the null hypothesis is that the difference in population means equals zero: $(\mu_{x,0} - \mu_{y,0}) = 0$. The logic is the following. If this null is rejected, we then have evidence that the population means are different.

Under the null hypothesis that $(\mu_{x,0} - \mu_{y,0}) = 0$, the t-statistic for the hypothesis test becomes:

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} + \frac{s_{\overline{y}}^2}{ny}}}$$

The numerator is the difference in sample means. The denominator is the standard error of the difference in sample means.

1.11 Application: Are Two Compact Cars Equally Fuel Efficient?

We have seen how to calculate power to test a hypothesis about a population mean. Often we want to compare differences in means. For example, suppose ATS wishes to compare the mileage of two different subcompact cars, A and B. The null hypothesis is that they are equally fuel efficient. An Effect Size of 2 MPG is chosen by ATS. Based on its prior experience, ATS expects the standard deviations, σ, of MPG for both vehicles to be 3. Recall that the cars being evaluated are subcompacts, so the standard deviation is less than for the SUV we studied above. ATS chooses the following for the design of its experiment:

- Significance level = .05
- Power = .85

We use the `tprstats.AB_nobs` command with significance level (.05), the standard deviation for a car of type A (3), the standard deviation for a car of type B (3), power (.85), and difference in means (2). The command calculates the number of observations needed for A and B as well as the total number of observations.

[77]:
```
tprstats.AB_nobs(diff_means=2, std1=3, std2=3, alpha=0.05, power=0.85, ratio=1)
```

[77]:
```
nobs1         41.386145
nobs2         41.386145
nobs_total    82.772291
dtype: float64
```

We obtain 82.77. This is the combined sample for the two vehicles. Multiplying by .5, we obtain 41.34. Rounding, we conclude that ATS needs 41 vehicles of each type. To be conservative, ATS might round up, and test 42 cars of each type. For practice, complete Exercise 2.

In the above command, we specify equal sample sizes by setting ratio=1. We can change the ratio of sample sizes by changing the value set for ratio. If it were much more costly to test one type of vehicle than another, we might want to use more of the less expensive vehicle and fewer of the expensive one.

Suppose it is more costly to obtain type B vehicles to test than it is to obtain type A vehicles. ATS might then set ratio = 2/3. This would specify that there would be 50 percent more cars of type A than type B. We then execute the following command:

[78]:
```
tprstats.AB_nobs(diff_means=2, std1=3, std2=3, alpha=0.05, power=0.85, ratio=2/
↪3)
```

[78]:
```
nobs1         51.682413
nobs2         34.454942
nobs_total    86.137355
dtype: float64
```

We obtain a total sample size of 86.13 with 34.45 type B vehicles and 51.68 type A vehicles. Of course, we cannot use a fraction of a car. Rounding, we obtain 35 vehicles of Type B and 52 of Type A. Thus, a total of 87 vehicles need to be tested. With unequal sample sizes, the total number of vehicles to be tested is higher (87) than with equal samples sizes (82). When the standard deviations are the same for the two types, unequal sample sizes will always require a larger total.

Note: We want both samples to be large enough to satisfy the central limit theorem, so, with unequal samples, we typically want to require that each sample have at least 25.

1.12 Application: Atlanta Water Conservation Experiment

The US Geological Survey has highlighted the increasing severity of droughts in the US. "Drought is a serious environmental threat across the United States. Climate change exacerbates droughts by making them more frequent, longer, and more severe."[2] Droughts in turn have created challenges for companies supplying water to residents of cities and towns. The following headline in the Washington Post **"Arizona city cuts off a neighborhood's water supply amid drought"** underlines the severity of the problem.[3]

We previously reviewed the experiment in water conservation conducted in Atlanta during a severe drought in 2007. This application is very instructive both for understanding experimental design and for appreciating the value of experiments.

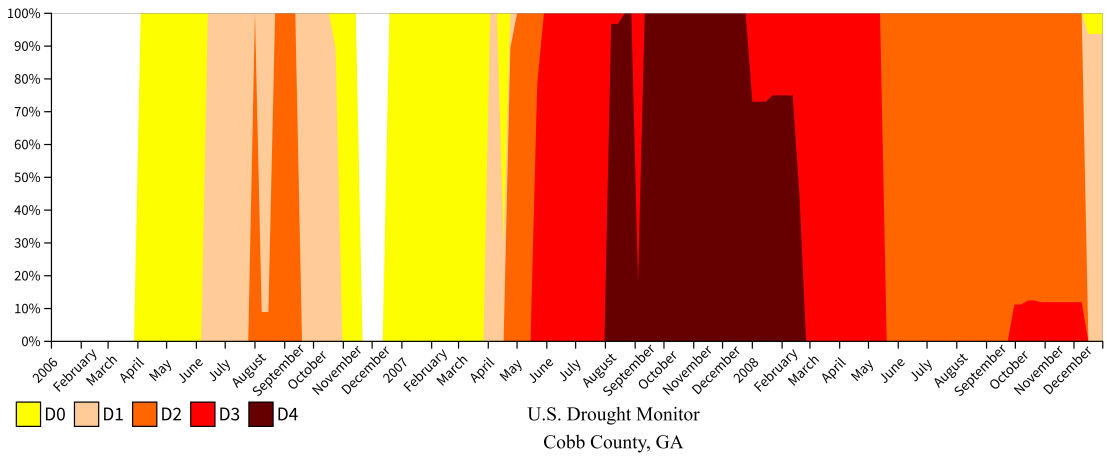
[79]:
```
Atlanta_Rainfall_1996_2023 = pandas.read_excel("data/Atlanta_Rainfall_1996_2023.
↪xlsx")
```

To display the severity of the drought, I provide two portrayals. The first, shown below shows severity of the drought in detail for 2006 through 2009.[4]

Color coding:
Yellow: Abnormally Dry
Beige: Moderate Drought
Orange: Severe Drought
Red: Extreme Drought
Brown: Exceptional Drought

If you are viewing a printout in black and white, the darker the shading, the more severe was the drought.

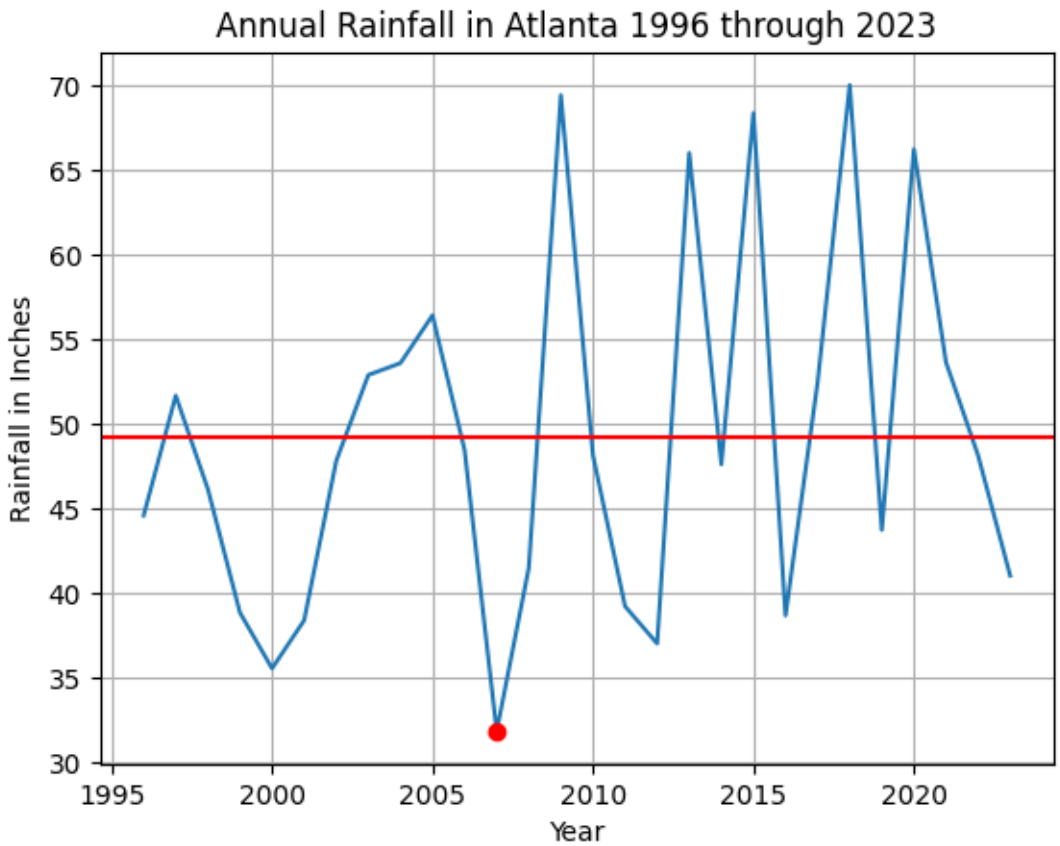
[2] https://www.usgs.gov/science/science-explorer/climate/droughts-and-climate-change
[3] https://www.washingtonpost.com/climate-environment/2023/01/16/rio-verde-foothills-water-scottsdale-arizona/
[4] Source: https://www.drought.gov/states/georgia/county/cobb



The second portrayal is the graph below that we reviewed previously showing rainfall in Atlanta from 1996 through 2023.[5]. Rainfall in 2007 is marked by a dot on the graph. The average annual rainfall over this period was 49.2 inches. This average is shown by the horizontal line on the graph. The drought in 2007 is quite evident on the graph. As you can see from the data file, rainfall in 2007 was only 31.85 inches. This was less than two thirds of the annual average.

[80]:
```
plt.plot(Atlanta_Rainfall_1996_2023["Year"],␣
↪Atlanta_Rainfall_1996_2023["Total"])
plt.axhline(y=49.2, color="red")
plt.xlabel("Year")
plt.ylabel("Rainfall in Inches")
plt.title("Annual Rainfall in Atlanta 1996 through 2023")
plt.plot(2007, 31.85, marker="o", color="red") # plots the red point at x=2007,␣
↪y=31.85
plt.grid(True)
```

[5]Source: https://www.weather.gov/ffc/rainfall_scorecard



[80]:

The overview that follows is from reports on the experiment by the team that designed and conducted the experiment, and evaluated the findings of the experiment.[6] I thank these authors for their generosity in sharing their data with us. As you will see, the experiment was conducted in Cobb County, which is in the Atlanta, Georgia Metropolitan area.

1.12.1 Objectives of the Experiment

"The Cobb County Water System (CCWS), an agency of the Cobb County Government, distributes treated surface water to about 170,000 customers. About 150,000 are residential customers that reside in single-family dwellings. CCWS obtains water from surface supplies that have been affected by periodic drought conditions since 1998. To reduce water use among its residential customers, the CCWS agreed to initiate a targeted, mail-based conservation education program in 2007 through

[6] Paul J. Ferraro, Juan Jose Miranda, and Michael K. Price, "The Persistence of Treatment Effects with Norm-Based Policy Instruments: Evidence from a Randomized Environmental Policy Experiment," *American Economic Review: Papers and Proceedings* 2011, 101:3, 318–322.
Paul J. Ferraro and Michael Price, "Using Nonpecuniary Strategies to Influence Behavior: Evidence from a Large-Scale Field Experiment," *Review of Economics and Statistics*, March 2013, 95(1): 64–73.
María Bernedo, Paul J. Ferraro, and Michael Price The Persistent Impacts of Norm-Based Messaging and Their Implications for Water Conservation Journal of Consumer Policy (2014) 37:437–452.

a randomized experimental design.

"Treatment 1, the technical advice treatment, was a technical advice treatment that provided households with an information only message: a two-sided "tip sheet" listing ways to most effectively reduce water use and whom to contact for more information.

"Treatment 2, the weak social norm treatment, augmented Treatment 1 by including a personally addressed letter to the household on official CCWS stationary. In addition to information found on the customer's month bill (and the tip sheet), the letter includes the following normbased appeal: Cobb County residents consume almost one of out every ten gallons of Georgia's public water supply. As a result, our water use has a large impact on the ability of Georgia's waterways to protect wildlife and dilute pollutants that threaten human health. We need your help. Act on the tips listed in the enclosed tip sheet. We all have to do our part to protect Cobb County's precious water resources. Reducing our water consumption today is important for preserving our environment and our economy for future generations. Please don't waste water. Remember: every drop counts!

"Treatment 3, the strong social norm treatment, augmented Treatment 2 with a comparison of the household's water use to the median county household use for period June to October 2006, and indicated the percentile in which the household fell during this period. This comparison read: As we enter the summer months, we thought that you might be interested in the following information about your water consumption last year: Your own total consumption June to October 2006: 52,000 gallons Your neighbors' average (median) consumption June to October 2006: 35,000 gallons. You consumed more water than 73 percent of your Cobb County neighbors."

1.12.2 Design of the Experiment

The experiment was designed as follows. The authors report that "…the CCWS required a minimum detectable treatment effect of a 2% decline in average water use and statistical power of at least 0.90." This tells us that the effect size was specified to be 2% of average water use, which was 725 gallons. Thus, the effect size we specified to be 725 gallons.

There were a total of 106,669 households in the experiment. Of these, 71,643 were assigned to the control group. Treatment groups 1 and 2 each had 11,675 households. Group 3 had 11,676 households.[7] Let's calculate the power for Treatment 1. For this, we use information about Treatment 1 and the control group. As note above, there were 71,643 households in the control group 11675 households in Treatment 1. The standard deviation of water usage was anticipated to be 22,500 for all groups.

The command for calculating power of this experiment is given below. Executing this command, we obtain power=.897. This rounds to .90 which was the power that CCWS requested for the experiment. The exact same calculation applied to Treatment 2 and Treatment 3. Hence, the experiment had a power of .90 for all three treatment groups.

[81]:
```
tprstats.AB_power(diff_means=725, nobs1=71643, nobs2=11665, std1=22500,␣
↪std2=22500, alpha=0.05)
```

[81]:
```
np.float64(0.897478595699841)
```

[7] The one extra household in group 3 is a result of having to allocate an integer number of households. Obviously, 1 additional household will be inconsequential to the results.

[81]:

```
```

In our ATS application to test fuel efficiency of two compact cars, we used `tprstats.Ab_nobs` to calculate sample sizes needed to achieve a given power. For our current analysis, we used `tprstats.AB_power` to calculate power when sample sizes are given.

1.12.3 Implementing the Experiment

Having completed the design of the experiment, as described above, the research team then set about implementing the experiment. The next step is to randomly assign households across the four groups. Here are the steps for doing so using Excel:

1. Put the account number of all households into Column A of an Excel spreadsheet. There are 106,669 households, so the spreadsheet will have 106,669 account numbers.
2. Draw 106,669 random numbers and put them into Column B of an Excel spreadsheet. You can draw a random number in Excel by entering =RAND() into the cell. So, put =RAND() into cell B2. Then drag down to cell 106670. You will then have 106,669 random numbers in column B.
3. Mark columns A and B and sort on column B. The account numbers will then be in random order.
4. Assign the first 11,675 households to treatment group 1, the next 11,675 to treatment group 2, and the next 11,675 to treatment group 3. Assign the remainder to the control group.

The next step is then to implement the treatments for the three treatment groups by sending out the appropriate letter to each household. No treatment is given to the control group.

When the experiment is completed, the data are analyzed to determine, for each treatment, whether the treatment significantly reduced water consumption. For this experiment, CCWS collected data for all households in the experiment for several years after the treatments were implemented.

1.12.4 Test The Null Hypothesis of No Effect of Treatment 3

We did this test earlier in the course. We briefly repeat the test here and then see how to do the test with regression. Import file Water_Conservation_Data.xlsx. For convenience, I renamed the file WatDat.

[82]:

```
WatDat = pandas.read_excel("data/Water_Conservation_Data.xlsx")
```

Select observations in GROUP 3. The following command selects the subset of WatDat for households in treatment group three.

[83]:

```
WatDat3 = WatDat.query("GROUP==3")
```

View mean, standard deviation, and number of observations for water use in summer 2007 for households in treatment group three.

[84]:

```
WatDat3["SUMMER_07"].describe()
```

[84]:

```
count    11676.000000
mean        34.857999
std         26.335250
```

```
min          0.000000
25%         19.000000
50%         27.000000
75%         42.000000
max        534.000000
Name: SUMMER_07, dtype: float64
```

Select observations in GROUP 4, the control group.

[85]:
```
WatDat4 = WatDat.query("GROUP==4")
```

View mean, standard deviation, and number of observations for group four in summer of 2007.

[86]:
```
WatDat4["SUMMER_07"].describe()
```

[86]:
```
count    71643.000000
mean        36.472537
std         29.251982
min          0.000000
25%         20.000000
50%         28.000000
75%         44.000000
max       1632.000000
Name: SUMMER_07, dtype: float64
```

Using the results above, we find the difference in means is: $(\overline{X} - \overline{Y}) = (34.858 - 36.472) = -1.614$. Water use is measured in thousands of gallons. This tells us that, water use per household in Treatment Group 3 was 1,614 gallons less than water use per household in the control group (group 4).

Is the above difference statistically significant? We calculate the t-statistic to test the null hypothesis that there is no difference in population means between Treatment Group 3 and the Control Group:

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{ny}}} = \frac{34.858 - 36.472}{\sqrt{\frac{26.335^2}{11676} + \frac{29.252^2}{71643}}} = -6.043$$

The p-value for this t-statistic is far below .05. Hence, the savings of 1,614 is statistically significant.

[87]:
```
2 * t.cdf(-6.043, (11676 + 71643 - 2))
```

[87]:
```
np.float64(1.5191391415639335e-09)
```

The above is good news! The savings per household of 1,614 gallons is substantial in magnitude and it is statistically significant. Be sure you understand how we obtained each number in the above calculation.

1.12.5 Understanding Heteroskedasticity

This application is very helpful for you in understanding heteroskedasticity. For convenience, the t-statistic above is repeated here:

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} + \frac{s_{\overline{y}}^2}{ny}}}$$

Note where the sample variances $s_{\overline{x}}^2$ and $s_{\overline{y}}^2$ appear in the formula. As you can see, the formula allows the variances to be different. In the water conservation experiment, the sample variances are indeed different: $\frac{34.858-36.472}{\sqrt{\frac{26.335^2}{11676}+\frac{29.252^2}{71643}}}$. In particular, $s_{\overline{x}}^2 = 26.335^2$ and $s_{\overline{y}}^2 = 29.252^2$. The estimated variance is higher for the control group than for treatment Group 3.

This nicely illustrates heteroskedasticity. Our calculation of the t-statistic above takes account of these unequal variances.

1.12.6 Using Regression To Test The Hypothesis

Regression is often used to do a test of differences in means. Regression gives the same result as obtained by calculating the t-statistic as we did above. When we have the data, regression is very convenient because it automates all of the calculations.

Using the query command, we select observations that are either in treatment 3 (GROUP==3) or the control group (GROUP==4). The vertical line | denotes "or".

[88]:
```
WatDat34 = WatDat.query("GROUP==3 | GROUP==4")
```

Suppose instead we used the following in the query command: `GROUP==3 & GROUP==4`.

What data would we extract? None. Each household was assigned to only one group. Hence, there is no household that appears in both groups 3 and 4.

Estimate a regression of SUMMER_07 vs TREAT3

[89]:
```
WatReg34 = tprstats.model("cs", "SUMMER_07~TREAT3", WatDat34)
```

View the regression results.

[90]:
```
WatReg34.summary()
```

[90]:

Dep. Variable:	SUMMER_07	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	0.000
No. Observations:	83319	**F-statistic:**	36.54
Covariance Type:	HC1	**Prob (F-statistic):**	1.50e-09

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	36.4725	0.109	333.730	0.000	36.258	36.687
TREAT3	-1.6145	0.267	-6.045	0.000	-2.138	-1.091

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Look at the results in the TREAT3 row in the above output. Then look back at the calculations that we did when we calculated the t-statistic "by hand". Be sure that you see where we calculated each item that appears in the TREAT3 row of this regressions. This demonstrates that regression is a very convenient way to estimate the treatment effect from the experiment and to test if the treatment effect was significantly different from zero. Below, additional detail is provided to help you better understand the regression. You will become more comfortable with this use of regression as we proceed in the course.

The regression summary does precisely the same calculation of the t-statistic as we did in the formula above, taking account of heteroskedasticity. We calculated a t-statistic=6.045. As you can see above, the regression output contains the same value for the t-statistic.

1.12.7 Use of an Indicator Variable in a Regression Equation

In the above regression, our explanatory variable is a (0,1) variable. TREAT3 takes on values of 0 and 1. Such a variable is called an **indicator variable**, and sometimes called a dummy variable. For households in the control group, TREAT3=0. Hence, the predicted value for such a household is the intercept, 36.4725. This predicted value is the same for all households in the control group; it is mean water usage for households in the control group, 36.4725 thousand gallons per household.

For households in treatment group 3, TREAT3=1. Hence, the predicted value is the same for all households in treatment group 3, (36.4725-1.6145)=34.858 thousand gallons per household. This is mean water usage of households in treatment group 3.

Using the above values, calculate the difference in water use between treatment group 3 and the control group and we obtain [(36.4725-1.6145)-36.4725]=-1.6145 thousand gallons per household. This demonstrates why the coefficient of TREAT3 is the treatment effect. This illustrates the convenience of using regression for evaluating the results of an experiment. In the row labeled TREAT3 in the regression we obtain the estimated treatment effect and the test of the null hypothesis that the population treatment effect is zero.

1.12.8 Was The Assignment to Treatment and Control Really Random?

When evaluating an experiment, it is good practice to investigate if the assignment to treatment and control was really random. If the assignment was random, then the treatment assignment should not predict a difference in water use in 2006 because water use in 2006 was determined before the experiment was conducted in 2007. The following regression estimates the difference in water usage in 2006 for treatment group 3 relative to the control group. The estimated difference is 137 gallons and the p-value=.736. This high p-value supports the claim that the assignment was indeed random. This is called a **balance check**. It is a way to persuade a skeptic who might doubt that assignment to treatment condition was really random.

[91]:
```
WatReg2006 = tprstats.model("cs", "WATER_2006~TREAT3", WatDat34)
WatReg2006.summary()
```

[91]:

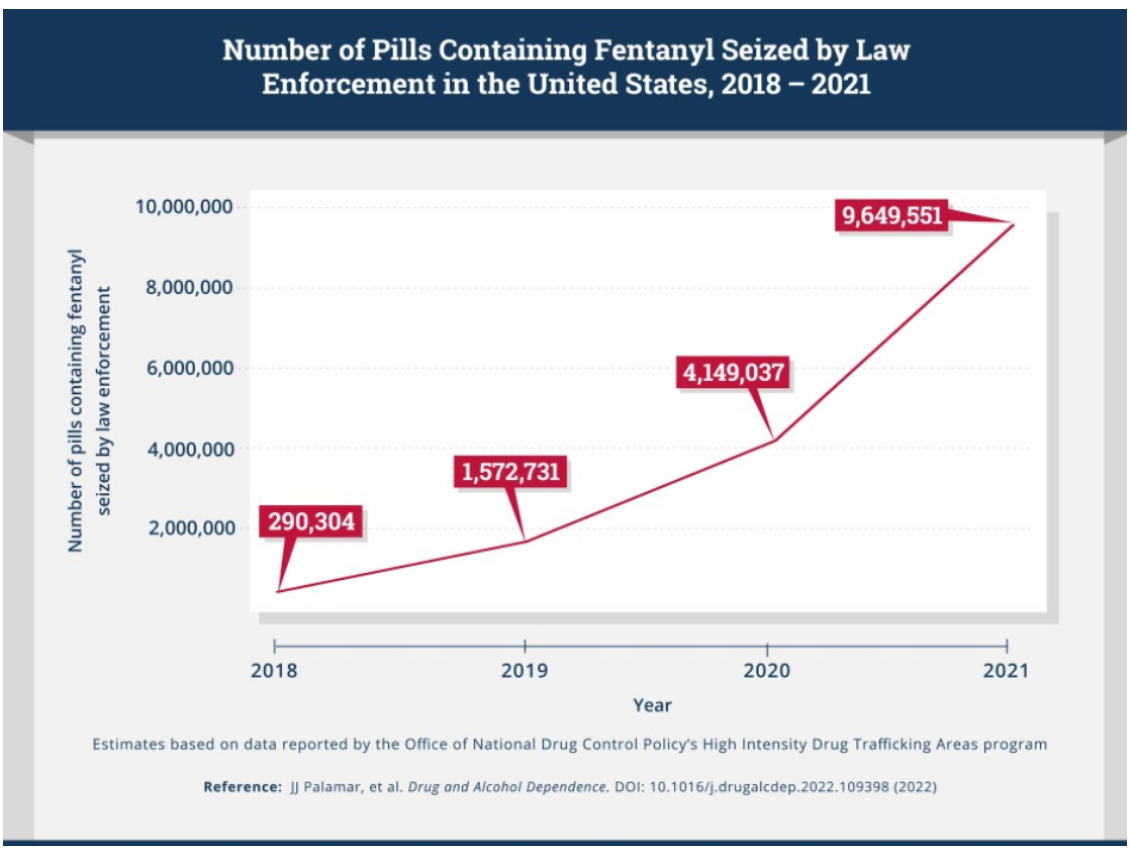
Dep. Variable:	WATER_2006	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	-0.000
No. Observations:	83319	**F-statistic:**	0.1140
Covariance Type:	HC1	**Prob (F-statistic):**	0.736

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	58.2982	0.155	377.068	0.000	57.995	58.601
TREAT3	0.1374	0.407	0.338	0.736	-0.660	0.935

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

1.13 Application: Effectiveness of Opioids vs Non-Opioids

Chapter 2 provided a brief overview of the opioid epidemic and the deplorable role of several major companies in propagating opioid use. By way of further background, the graph below shows the increase in seizure by law enforcement of pills containing fentanyl.

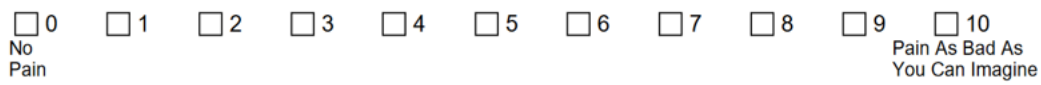


The US Center for Disease Control (CDC) identified three waves of opioid deaths. Regarding the first wave, the CDC concluded: "The first wave began with increased prescribing of opioids in the 1990s, with overdose deaths involving prescription opioids (natural and semi-synthetic opioids and methadone) increasing since at least 1999."

https://www.cdc.gov/drugoverdose/data/analysis.html

In Chapter 2, we reviewed two experiments designed to test whether opioids are more effective for non-opioids for pain relief. In one of the experiments summarized in Chapter 2, we saw that the experimenters recruited patients from Veterans Affairs primary care clinics. Eligible patients had moderate to severe chronic back pain or hip or knee osteoarthritis pain despite analgesic use. There were 238 participants. The study randomly assigned 119 of these individuals to receive opioids for their pain and 119 to receive non-opioids for their pain. The experiment lasted for twelve months. The experimenters stated that "The primary outcome was pain-related function (Brief Pain Inventory [BPI] interference scale) over 12 months and the main secondary outcome was pain intensity (BPI severity scale). For both BPI scales (range, 0-10; higher scores=worse function or pain intensity), a 1-point improvement was clinically important."

Recall that pain intensity is measured as follows. A patient is asked to indicate pain intensity on an 11-point scale with 0 being no pain and 10 being the worst imaginable pain. This is called a Brief Pain Inventory (BPI) scale, and is shown below:



Below in bold face type is the summary of the experimental design as provided by the authors. Be sure that you see that this admirable summary tells us the value of the four items required to determine sample size for the experiment.

- What power was chosen?
- What effect size was chosen?
- What significance level was chosen?
- What standard deviation was chosen for each group?

Statistical Analysis

Assuming a 2-sided α level of .05 and a standard deviation of 2.7,[30] 115 patients completing the study per group were required for 80% power to detect a 1-point between-group difference in mean BPI interference at 12 months.

Practice: Using Python, do the calculation to confirm the number of subjects needed was 115 in each group. Use the information presented immediately above. Hint: You need to use **tprstats.AB_nobs()** to do the calculation. Please try doing this before looking at the solution below.

Participants in an experiment are often called **subjects**, so I will use that terminology in this discussion. The above summary correctly states that 115 subjects were required for each group in order to achieve a power of .80. The experiment actually had 119 individuals in each group. The use of 119 subjects for each group rather than 115 for each group was probably to allow for possible attrition. Attrition might arise, for example, if a subject moved to a new address without informing the experimenters.

Experimental outcomes were as follows. At 12 months, mean BPI pain severity was 4.0 in the opioid group (SD, 2.0) vs 3.5 in the non-opioid group (SD, 1.9). SD denotes standard deviation. Remember that lower BPI is better, so the above tells us that the non-opioid outcome was better than the opioid outcome!

In Chapter 2, we investigated whether this difference significant. The analysis is briefly repeated here. The information provided above permits us to do a test of differences in means using the formula presented previously in this chapter. For the non-opioid group $\overline{X} = 3.5$. For the opioid group $\overline{Y} = 4.0$. The sample sizes are n_x=119 for each group. The sample standard deviations were $s_{\overline{x}}$2=1.9 for the non-opioid group and $s_{\overline{y}}$=2.0 for the opioid group.

The null hypothesis is that the population mean is the same for the two treatments. Using the results above, we calculate the t-statistic as follows:

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} - \frac{s_{\overline{y}}^2}{ny}}} = \frac{(3.5 - 4.0)}{\sqrt{\frac{1.9^2}{119} + \frac{2^2}{119}}} = \frac{-.5}{.283} = -1.98$$

We have 119 individuals in each group. Hence, the degrees of freedom is $2 * 119 - 2 = 236$. Calculating the p-value, we obtain .049. This p-value is less that α=.05.

[92]:
```
2 * t.cdf(-1.98, 236)
```

[92]:
```
np.float64(0.048865256977707064)
```

Thus, we reject the null hypothesis that the two were equally effective in reducing pain severity and conclude that **non-opioids** were significantly **more** effective than opioids in reducing pain severity!

Another experiment on opioids for pain relief was conducted in hospital emergency rooms. The study found that an ibuprofen-Tylenol combination was as effective as opioids.[8]

Thus, two experiments on two very different types of pain, chronic pain and pain due to injuries, find that opioids are no better than non-opioids.

We will use the `tprstats.AB_nobs` command to calculate sample size for testing a difference in population means in the experiment to evaluate effectiveness of opioids relative to non-opioids for treating chronic pain. As presented above, the Brief Pain Inventory (BPI) is an 11 point scale. The effect size, denoted diff_means, was chosen to be 1 unit on the BPI. In this application, ratio=1. This tells Python that we want an equal number of subjects in each of the two samples. The anticipated standard deviation for population A was std1=2.7, the anticipated standard deviation for population B was std2=2.7, the significance level is .05 and power=.8. We do not need to specify the significance level because .05 is the default significance level used in AB_nobs. To specify a different significance level, you would include `alpha=a` in the command, where a is your desired level.

[93]:
```
tprstats.AB_nobs(diff_means=1, std1=2.7, std2=2.7, power=0.8, ratio=1)
```

[8] https://www.nytimes.com/2017/11/08/well/live/alternatives-to-opioids-for-pain-relief.html

[93]:
```
nobs1         115.404644
nobs2         115.404644
nobs_total    230.809289
dtype: float64
```

We see that 230.8 subjects are needed, with half, 115.4, assigned to each alternative. Of course, the number of subjects must be an integer, so the calculation tells us that 115 subjects are needed for each group. The experiment was done with 119 subjects in each group. The researchers added four extra subjects to each group to allow for possible dropouts.

Practice: How large should the sample be if desired power is .85?

The experiments on opioids versus non-opioids provide **causal** evidence that opioids are not better than non-opioids for two different kinds of pain. Even prior to the experiments, the dangers of opioid prescription and the doubtful advantage of opioids were laid out with impressive care and detail in a 2016 investigation by the Center for Disease Control.[9] A key paragraph from that report is reproduced here.

"In summary, evidence on long-term opioid therapy for chronic pain outside of end-of-life care remains limited, with insufficient evidence to determine long-term benefits versus no opioid therapy, though evidence suggests risk for serious harms that appears to be dose-dependent. These findings supplement findings from a previous review of the effectiveness of opioids for adults with chronic non-cancer pain. In this previous review, based on randomized trials predominantly $\leq$ 12 weeks in duration, opioids were found to be moderately effective for pain relief, with small benefits for functional outcomes; although estimates vary, based on uncontrolled studies, a high percentage of patients discontinued long-term opioid use because of lack of efficacy and because of adverse events (10)."

Had the careful work summarized in the previous paragraph been more widely heeded, the great toll on human life and health from excessive opioid prescription could have been greatly reduced. This work also demonstrates that careful statistical analysis can be exceedingly valuable in guiding decision making even in the absence of experiments. Indeed, as a manager, you will often be called upon to make decisions in circumstances where it is not feasible to conduct an RCT, or the decision must be made on a timetable that does not permit conducting an RCT. As we proceed in the course, we will study methods for statistical decision making that utilize data obtained in the course of normal business and organizational activities.

I should emphasize that I am not a physician. There may well be circumstances where opioids should be prescribed instead of non-opioids. As yet, however, I have not found any RCTs demonstrating superiority of opioids to non-opioids for pain relief.

1.13.1 Calculate Power Given Sample Size

You may sometimes read information about an experiment when power is not reported. You can do your own calculation using the AB_power command. As an example, suppose the researchers had chosen 140 subjects for each medication, a total of 280 subjects. We see that, with 280 subjects, power is .87.

[9] https://www.cdc.gov/mmwr/volumes/65/rr/rr6501e1.htm

[94]:
```
tprstats.AB_power(diff_means=1, nobs1=140, nobs2=140, std1=2.7, std2=2.7,␣
↪alpha=0.05)
```

[94]:
```
np.float64(0.8703517709271432)
```

1.14 Summary

This chapter has covered methods for **experimental design** for a test of a hypothesis about a single mean and for a test of a hypothesis of a difference in means. The chapter also presents several applications to familiarize you with the application of these methods and the analysis of data obtained from conducting experiments. Please return to the learning objectives at the beginning of this chapter to help guide your study as you seek to master the material.

1.15 Practice Exercises

1.15.1 Exercise 1

Suppose the manager of ATS adopts a standard of .85 for testing a hypothesis about the population mean for the luxury SUV. Please confirm, by modifying the appropriate command, that 58 vehicles would need to be tested.

1.15.2 Exercise 2

Consider the ATS evaluation of the fuel efficiency of the two economy cars. Suppose the standard deviation of fuel economy for both of the vehicles is 2.75. Please confirm, by modifying the appropriate command, that 35 vehicles of each type would would need to be tested. Explain intuitively in one sentence why fewer automobiles are needed than when the standard deviation for each vehicle is 3.

1.15.3 Exercise 3

Using the water conservation data set, test the null hypothesis that treatment 2 had no effect. Follow the steps that we used above to test null hypothesis that treatment 3 had no effect. Import the data set. Then select the observations for Group 2. Calculate the sample mean and sample standard deviation for Summer_07 for Group 2. Calculate how many observations are in Group 2 using NROW. With that information and the results above for Group 4 (the control group) calculate the difference in sample means and record the result. You should obtain -1.045. State verbally the interpretation of this result in one sentence. Calculate the standard deviation for Group 2. Then calculate and record the t-statistic. You should obtain a t-statistic of -3.70. Find and record the p-value for the t-statistic. Was the treatment effect significant at the 5% level?

[95]:
```
WatDat2 = WatDat.query("GROUP == 2") # Select data for treatment group 2.
xbar2 = WatDat2["SUMMER_07"].mean()  # Calculate mean for treatment group 2
xbar4 = WatDat4["SUMMER_07"].mean()  # We Previously selected data for group 4.␣
↪This calculates mean for group 4.
std2 = WatDat2["SUMMER_07"].std()    # Calculate standard deviation for␣
↪treatment group 2.
n2 = WatDat2["SUMMER_07"].size       # Calculate sample size for treatment␣
↪group 2.
```

```
n4 = WatDat4["SUMMER_07"].size       # Calculate sample size for treatment␣
↪group 4.
std4 = WatDat4["SUMMER_07"].std()    # Calculate standard deviation for␣
↪treatment group 4.
treat2effect = xbar2 - xbar4         # Calculate difference in means between␣
↪treatment group 2 and control group.
tstat2 = treat2effect / (std2**2 / n2 + std4**2 / n4) ** 0.5  # Calculate␣
↪t-stat to test the null hypothesis
                                                              # that the␣
↪difference in populations means equals zero.
pval2 = 2 * t.cdf(tstat2, (n2 + n4 - 2)) # Calculate p=value to test the null␣
↪hypothesis.

tstat2, pval2      # Report tstat and p-value
```

[95]: (np.float64(-3.701209103626142), np.float64(0.00021471265234063167))

Next, select observations for Groups 2 and 4 and put the results in a data file named WatDat24. Then, using this data file, estimate a regression with SUMMER_07 as the dependent variable and TREAT2 as the independent variable. Name the regression WatReg24. Compare the results of the regression output to the calculations you did in the previous paragraph and confirm that the results are the same.